

SIMATS ENGINEERING
ASSESSMENT TOOL 3 - LOW (CAPSTONE PROJECT)

COURSE CODE: CSA0927

**COURSE NAME: Programming in
Java**

COURSE OUTCOME COVERED

CO5: *Construct database-driven web applications by implementing JDBC connectivity to perform SQL operations such as insert, update, and delete using different statement types. (BL3,BL6)*

Assessment Tool Weightage: 40%

QUESTIONS: 1

Total Marks: 100

ANNEXURE A

CAPSTONE PROJECT

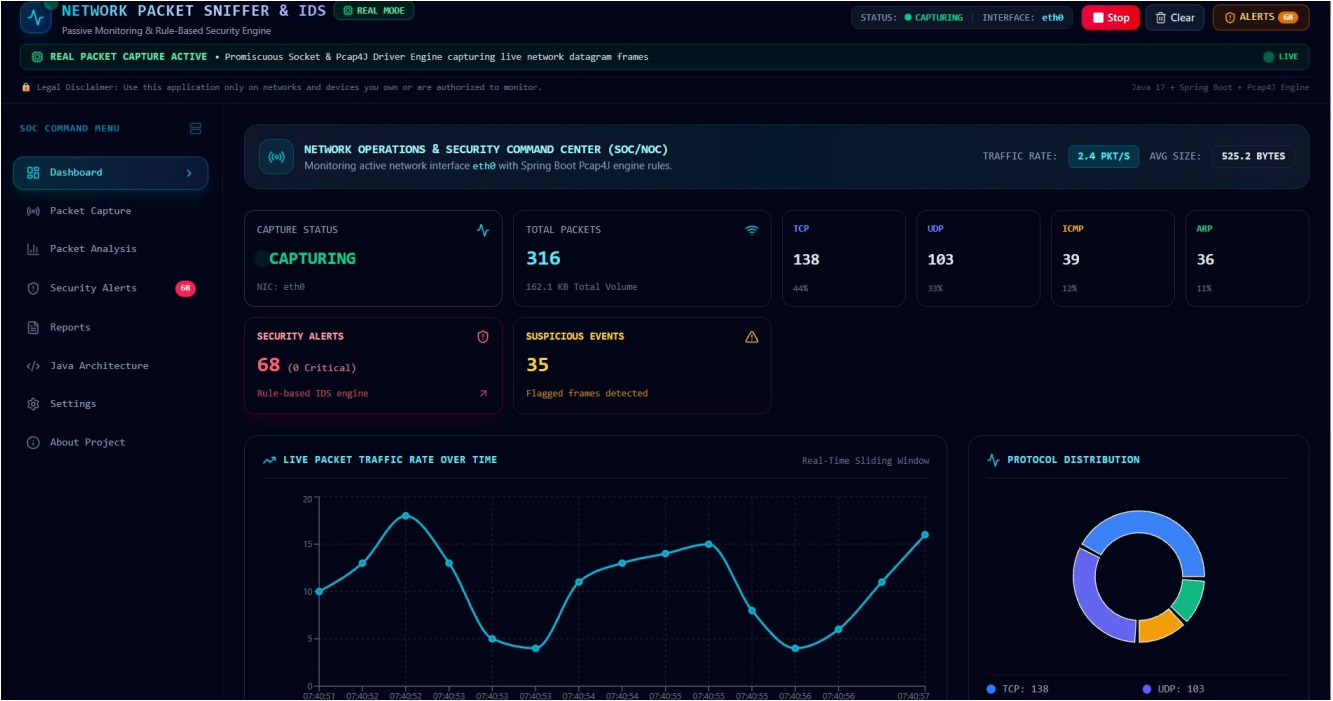
Code of Conduct:

I M.Mahammad Rafi (192421364)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M.MD.Rafi

DEMO PHOTOS:



NETWORK PACKET CAPTURE ENGINE							INTERFACE: Intel(R) Ethernet Connection I219-V (192.168.1.105)	
PROMISCUOUS SOCKET PACKET SNIFFER							Pcap4J / Npcap Java Engine	
STOP CAPTURE CLEAR EXPORT							PACKET COUNT: 316	STATUS: CAPTURING
PROTO: ALL TCP UDP ICMP ARP							Search IP, Port, or Protocol...	
#	TIME	SOURCE IP	SRC PORT	DESTINATION IP	DST PORT	PROTOCOL	PACKET LENGTH	
316	07:40:57.444	185.199.108.153	41431	192.168.1.105	500	UDP	388 Bytes	
315	07:40:57.044	142.250.190.46	56541	192.168.1.1	3389	TCP	480 Bytes	
314	07:40:56.643	10.0.0.42	20747	185.199.108.153	5353	UDP	88 Bytes	
313	07:40:56.244	192.168.1.20	-	192.168.1.1	-	ICMP	100 Bytes	
312	07:40:55.842	192.168.1.20	7144	185.199.108.153	8080	TCP	745 Bytes	
311	07:40:55.443	10.0.0.1	-	192.168.1.1	-	ARP	364 Bytes	

NETWORK PACKET VISUALIZATION DASHBOARD

TOTAL PACKETS

316

TCP PACKETS

138

UDP PACKETS

103

ICMP PACKETS

39

ARP PACKETS

36

AVG FRAME SIZE

525.2 B

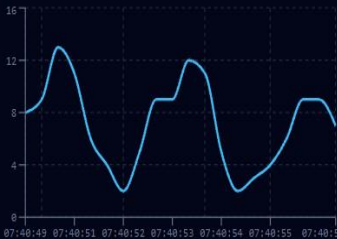
TRAFFIC RATE

2.4 P/s

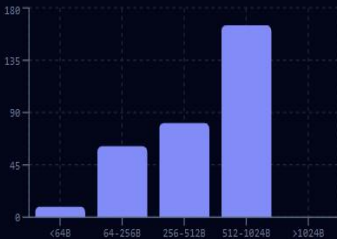
1. PROTOCOL DISTRIBUTION



2. TRAFFIC RATE OVER TIME



3. PACKET SIZE ANALYSIS



SECURITY INCIDENT & THREAT DASHBOARD

CLEAR ALL ALERTS

TOTAL ALERTS

68

CRITICAL ALERTS

0

HIGH ALERTS

68

MEDIUM ALERTS

0

LOW ALERTS

0

ACTIVE INTRUSION DETECTION RULES IN JAVA ENGINE:

1. PORT SCAN

>8 ports probed / 5s

2. HIGH TRAFFIC

>25 pkts/s from host

3. ICMP FLOOD

Burst ping flood

4. REPEATED CONN

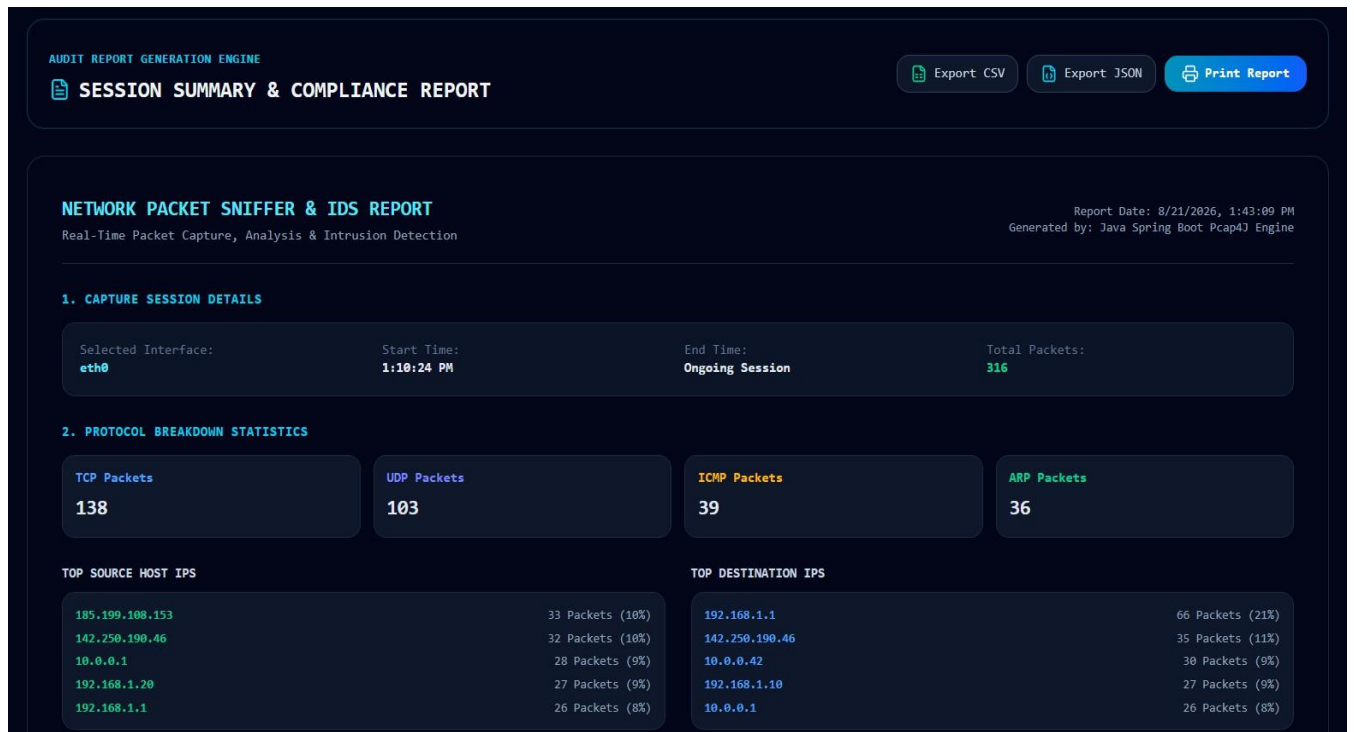
Rapid reconnect burst

5. SUSPICIOUS PORT

Ports 22, 23, 3389, 445...

SEVERITY FILTER: ALL CRITICAL HIGH MEDIUM LOW

ALERT ID	TIME	SEVERITY	SOURCE IP	DEST IP	DETECTION TYPE	DESCRIPTION	STATUS	ACTIONS
ALT-Y96XN2	07:40:55.842	HIGH	192.168.1.20	185.199.108.153	SUSPICIOUS_PORT	Connection attempt detected on sensitive po...	NEW	



GEO TAG PHOTO:



1. RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developin g	Level 1 – Beginning
Problem Statement & Innovation	20	Clearly defines the problem and presents an innovative, practical solution.	Defines the problem with a relevant solution.	Problem is partially clear with limited innovation.	Problem statement is unclear or irrelevant.
Technical Implementation	30	All modules are fully functional, integrated, and technically accurate.	Most modules function correctly with minor issues.	Core functionality is partially implemented.	Implementation is incomplete or incorrect.

Presentation & Communication	20	Well-organized, confident delivery with effective visuals and time management.	Clear presentation with good organization.	Presentation lacks clarity or consistent organization.	Poorly organized and difficult to follow.
Demonstration & Results	30	Successful live demonstration with accurate, validated results.	Demonstration works with minor functional issues.	Partial demonstration with limited validation.	Demonstration fails or results are incorrect.